

* Unit 3 Fourier Series *

* Periodic Function A function $f(x)$ is said to be periodic if its Time period 'T' if $f(x+T) = f(x)$

Ex $\sin x$ is periodic function, $T = 2\pi$ as well as $\cos x$
 $\cos x$ is periodic function, $T = 2\pi$ as well as $\sec x$
 $\tan x$ is periodic function, $T = \pi$ as well as $\cot x$

$$\sin 2x \Rightarrow \text{T.P.} = \frac{2\pi}{2} \Rightarrow \pi/2$$

$$\sin nx \Rightarrow \text{T.P.} = \frac{2\pi}{n}$$

* Even & Odd function

if $f(-x) = f(x) \Rightarrow$ even function

if $f(-x) = -f(x) \Rightarrow$ odd function

otherwise none (neither odd, nor even) function.

Ex $x, x^3, x^5, \dots, x^{2n+1}$ odd function
 $x^2, x^4, x^6, \dots, x^{2n}$ even function.

$\sin x, \tan x \Rightarrow$ odd function

$\cos x \Rightarrow$ even function

$e^x, x+x^2 \Rightarrow$ none

* Property

$$\int_{-a}^a f(x) dx = \begin{cases} 0 & f(x) \text{ is odd;} \\ 2 \int_0^a f(x) dx & f(x) \text{ is even;} \end{cases}$$

$\rightarrow$ Odd functions are always symmetric about origin.

$\rightarrow$ even functions are symmetric about Y-axis.

* Fourier Series:-

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx] \quad \text{--- (i)}$$

$$* \quad c < x < c+2\pi \quad \left[\text{Generally } c = 0 \text{ or } -\pi \right]$$

where

$$a_0 = \frac{1}{2\pi} \int_c^{c+2\pi} f(x) dx ; \quad \text{--- (ii)} \quad (n \in \mathbb{I})$$

$$a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx ; \quad \text{--- (iii)}$$

$$b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx ; \quad \text{--- (iv)}$$

Formule (2), (3) & (4) known as Euler's formula
[RTU - 2022]

put $c = -\pi$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

Note for limit $-\pi$ to $+\pi$

is if $f(x)$ is even function

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

then $b_n = 0$;

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

(ii) if $f(x)$ is odd function

then $a_0 = 0, a_n = 0$;

Q Find Fourier series of $f(x) = |x|$, $-\pi < x < \pi$

Hence show that

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \sin nx dx$$

$f(x) = |x|$ (even function)

($b_n = 0$)

$b_n = 0$

Solⁿ

$$\text{So } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| dx$$

$$a_0 = \frac{2}{2\pi} \int_0^{\pi} (x) dx$$

$$a_0 = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi}$$

$$a_0 = \frac{1}{2\pi} [\pi^2 - 0]$$

$$a_0 = \frac{\pi}{2} \quad (\text{RTU 2024})$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \cos nx dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx dx$$

$$a_n = \frac{2}{\pi} \left[x \int_0^{\pi} \cos nx dx - \int_0^{\pi} \frac{d(x)}{dx} \cdot \int_0^{\pi} \cos nx dx \right]$$

-sin nx

$$a_n = \frac{2}{\pi} \left[\left[\frac{x \sin nx}{n} \right]_0^{\pi} + \left[\frac{\cos nx}{n^2} \right]_0^{\pi} \right]$$

$$a_n = \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx - \cos 0}{n^2} \right]$$

$$a_n = \frac{2}{\pi} \left[0 \frac{\cos n\pi - 1}{n^2} - \frac{(\cos n\pi - 1)}{n^2} \right]$$

$$a_n = \frac{-4}{\pi n^2} \quad a_n = \frac{2}{n^2 \pi} (\cos n\pi - 1)$$

$$a_n = \begin{cases} \frac{-4}{\pi n^2} ; & n \rightarrow \text{odd} \\ 0 ; & n \rightarrow \text{even} \end{cases}$$

Now the series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

Ans:- For n even

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} [\cos nx - 1] \cos nx + 0$$

$$f(x) = \frac{\pi}{2} + \left[-\frac{4}{\pi} \cos x + 0 - \frac{4}{\pi 3^2} \cos 3x + \frac{\cos 5x}{\pi 5^2} - \dots \right]$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right] \quad \underline{\text{Answer}}$$

Now put $x=0$

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

Hence Proved

Q Find the fourier series of the function

$$f(x) = x^2, \quad -\pi < x < \pi$$

Hence Show that

$$\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \dots \quad (\text{RTU-2023})$$

Soln Given $f(x) = x^2$ (even function)

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi} \left[\frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{\pi^2}{3}$$

$$b_n = \int_{-\pi}^{\pi} x^2 \sin nx dx = 0$$

Now for

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx \, dx$$

$\downarrow$ $\downarrow$
 I II

by by-parts

$$a_n = \frac{2}{\pi} \left[\int_0^{\pi} x^2 \cos nx \, dx - \int_0^{\pi} \frac{d(x^2)}{dx} \int_0^{\pi} \cos nx \, dx \right]$$

$$a_n = \frac{2}{\pi} \left[\left[x^2 \left(\frac{\sin nx}{n} \right) \right]_0^{\pi} - \int_0^{\pi} \frac{2x \sin nx}{n} \, dx \right]$$

$$a_n = \frac{2}{\pi} \left[\left[\frac{x^2 \sin nx}{n} \right]_0^{\pi} - 2 \left[\left(\frac{x \cos nx}{n^2} \right) + \int_0^{\pi} \frac{\sin nx}{n^2} \, dx \right] \right]$$

$$a_n = \frac{2}{\pi} \left[0 - 2 \left[\left(\frac{x \cos nx}{n^2} \right) + \left(\frac{\sin nx}{n^3} \right) \right]_0^{\pi} \right]$$

$$a_n = \frac{2}{\pi} \left[-2 \left[\frac{-x \cos nx}{n^2} - 0 \right] + 0 \right]$$

$$a_n = \frac{2}{\pi} \left[+ \frac{2x \cos nx}{n^2} \right] \quad \star \left[\begin{array}{l} \text{Note } \sin nx = 0 \\ \cos nx = (-1)^n \end{array} \right] \star$$

$$a_n = \frac{4}{n^2} \cos n\pi$$

Now the fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cdot \cos nx$$

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nx \cos nx$$

$$x^2 = \frac{\pi^2}{3} + 0 - 4\cos x + \cos 2x - \frac{4}{3^2} \cos 3x + \frac{4}{4^2} \cos 4x - \dots$$

$$x^2 = \frac{\pi^2}{3} - \frac{4\cos x}{1^2} + \frac{4\cos 2x}{2^2} - \frac{4}{3^2} \cos 3x + \frac{4}{4^2} \cos 4x \dots$$

Now put $x=0$

$$\Rightarrow 0 = \frac{\pi^2}{3} - \frac{4}{1^2} + \frac{4}{2^2} - \frac{4}{3^2} + \frac{4}{4^2} - \frac{4}{5^2} + \frac{4}{6^2} - \dots$$

$$\Rightarrow \frac{\pi^2}{3} \times \frac{1}{4} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \frac{1}{6^2} - \dots$$

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \dots$$

Hence Proved

Q Find the Fourier Series of the function :

$$f(x) = \begin{cases} x+x & ; -\pi < x < 0 \\ x-x & ; 0 < x < \pi \end{cases}$$

for the interval $-\pi < x < \pi$

Soln for limit $(-\pi, 0)$ $f(x) = x+x$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^0 (x+x) dx$$

$$a_0 = \frac{1}{2\pi} \left[\pi x + \frac{x^2}{2} \right]_{-\pi}^0 \Rightarrow \frac{1}{2\pi} \left[\pi^2 + \frac{\pi^2}{2} \right] \Rightarrow \frac{3\pi}{4}$$

for limit $(0, \pi)$ $f(x) = \pi - x$

$$a_0 = -\left(\pi^2 - \frac{\pi^2}{2}\right) \frac{1}{2\pi} \quad (0, \pi)$$

$$a_0 = -\frac{\pi}{4}$$

Now Complete.

$$a_0 = \frac{3\pi}{4} + \frac{\pi}{4}$$

$$a_0 = \pi/2$$

Now for

$$a_n = \frac{1}{\pi} \int_{-\pi}^0 (\pi+x) \cos nx \, dx + \frac{1}{\pi} \int_0^{\pi} (\pi-x) \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \left[\frac{\pi \sin nx}{n} + \frac{x \cos nx}{n} + \frac{\cos nx}{n^2} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{\pi \sin nx}{n} - \frac{x \cos nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$a_n = \frac{1}{\pi} \left[\frac{1}{n^2} - \frac{\cos n\pi}{n^2} \right] + \frac{1}{\pi} \left[-\frac{\cos n\pi}{n^2} + \frac{1}{n^2} \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{2}{n^2} - 2 \frac{\cos n\pi}{n^2} \right]$$

$$a_n = \frac{2}{n^2\pi} [1 - \cos n\pi] \underline{A_2}$$

$$a_n = \begin{cases} \frac{4}{n^2\pi} & n \rightarrow \text{odd} \\ 0 & n \rightarrow \text{even} \end{cases}$$

$$\text{for } b_n = \frac{1}{\pi} \int_{-\pi}^0 (\pi+x) \sin nx dx + \frac{1}{\pi} \int_0^{\pi} (\pi-x) \sin nx dx$$

$$b_n = \frac{1}{\pi} \left[\frac{-\pi \cos nx}{n} + \right. \\ \left. \right. \quad \quad \quad (\text{odd})$$

$$b_n = 0$$

So Fourier Series

$$f(x) = \frac{\pi}{2} + \frac{4 \cos x}{1^2 \pi} + 0 + \frac{4 \cos 3x}{3^2 \pi} + \frac{4 \cos 5x}{5^2 \pi} \dots$$

$$f(x) = \frac{\pi}{2} + \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} \dots \right]$$

* Half range Series:-

When the fourier series is required in the half interval that is $0 < x < \pi$, not in full interval, we can assume $f(x)$ to be even function or odd function in interval $-\pi < x < \pi$.

If function $\rightarrow$ even $\Rightarrow b_n = 0 \Rightarrow$ Cosine series
function $\rightarrow$ odd $\Rightarrow a_n = a_0 = 0 \Rightarrow$ Sine series

$f^n \rightarrow$ even

$$b_n = 0$$

$$f(x) = \sum_{n=1}^{\infty} a_n \cos nx + a_0$$

Cosine series

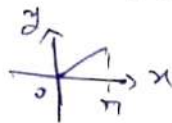
$f^n \rightarrow$ odd

$$a_0 = a_n = 0$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

Sine series

Q Find the fourier sine & cosine series of $f(x) = x$ in interval $0 < x < \pi$. $\rightarrow$



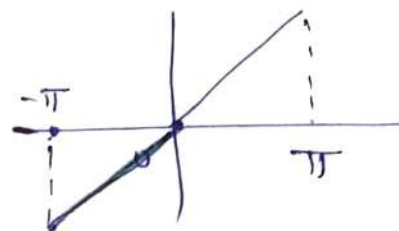
Soln $f(x) = x$ (odd function)

① Sine series $\Rightarrow$

$$f(x) = x ; -\pi < x < \pi$$

$$a_0 = 0, a_n = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx dx$$



(odd function)

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx \, dx$$

$\downarrow$ $\downarrow$
 I II

$$\Rightarrow b_n = \frac{1}{\pi} \left[x \int \sin nx \, dx - \int \frac{d(x)}{dx} \int \sin nx \, dx \right]_{-\pi}^{\pi}$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[-x \frac{\cos nx}{n} + \int 1 \cdot \frac{\cos nx}{n} \, dx \right]_{-\pi}^{\pi}$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[-x \frac{\cos nx}{n} + \frac{\sin nx}{n^2} \right]_{-\pi}^{\pi}$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[-\pi \frac{\cos n\pi}{n} + (-\pi) \frac{\cos n\pi}{n} \right]$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[-\frac{2\pi \cos n\pi}{n} \right]$$

$$\Rightarrow b_n = -\frac{2}{n} \cos n\pi$$

So Fourier series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$f(x) = \sum_{n=1}^{\infty} -\frac{2}{n} \cos n\pi \sin nx$$

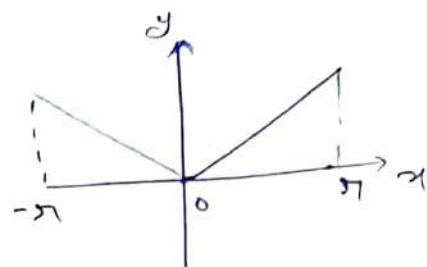
$$f(x) = 2 \sin x - \frac{2}{2} \sin 2x + \frac{2}{3} \sin 3x - \frac{2}{4} \sin 4x + \frac{2}{5} \sin 5x$$

$$f(x) = 2 \sin x - \sin 2x + \frac{2}{3} \sin 3x - \frac{\sin 4x}{2} + \frac{2}{5} \sin 5x - \dots$$

② Now for Cosine Series

function should be even

$$f(x) = |x| \quad -\pi < x < \pi$$



$$b_n = 0$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| dx$$

$$a_0 = \frac{1 \times 2}{2\pi} \int_0^{\pi} x dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi}$$

$$\boxed{a_0 = \pi/2}$$

$$\text{for } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \cos nx dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx dx$$

$$a_n = \frac{2}{\pi} \left[x \int \cos nx dx - \int \frac{d(x)}{dx} \int \cos nx dx \right]_0^{\pi}$$

$$a_n = \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$a_n = \frac{2}{\pi} \left[0 + \frac{\cos n\pi}{n^2} - \frac{\cos(0)}{n^2} - 0 \right]$$

$$a_n = \frac{2}{n^2} (\cos n\pi - 1)$$

So series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$f(x) = \frac{\pi}{2} + \frac{(-4)}{\pi} \cos x + 0 - \frac{4}{3^2 \pi} \cos 3x + 0 - \frac{4}{5^2 \pi} \cos 5x \dots$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$$

* Change of Interval:-

$$(-l < x < l)$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right]$$

("General formula for fourier series")

where, $a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

Q Find half range cosine series of function $f(x) = (2x-1)$ in $0 < x < 1$ [RTU 2023]

Solⁿ Given function

$$f(x) = (2x-1) \quad 0 < x < 1$$

for cosine series, we assume even function.

$$f(x) = |2x-1|$$

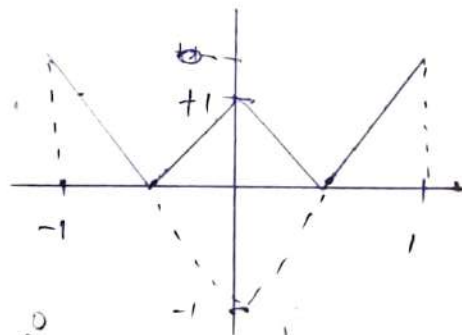
by half range series -

$$a_0 = \frac{1}{2l} \int_{-l}^l f(x) dx$$

$$a_0 = \frac{1}{2(1)} \int_{-1}^1 |2x-1| dx = \frac{1}{2} \left[-\int_{-1}^0 (2x-1) dx + \int_0^1 (2x-1) dx \right]$$

$$a_0 = \frac{1}{2} \int_{-1}^1 (2x-1) dx = \frac{1}{2} \left[x^2 - x \right]_{-1}^1 + \frac{1}{2} \left[x^2 - x \right]_{-1}^1$$

$$= \frac{1}{2} (-0 + 0 - (-1 - 1)) = \frac{1}{2} (2) = 1$$



$$a_0 = \frac{2}{2} \left[\frac{2x^2}{2} - x \right]_0^1$$

$$\int_{-1}^1 (2x+1) dx = \left[x^2 + x \right]_{-1}^1$$

$$a_0 = \frac{2}{2} [1 - 1 - (-1) + (1)]$$

$$\int_{-1}^1 (2x+1) dx = \left[x^2 + x \right]_{-1}^1 = 1 + 1 - (-1 - 1) = 4$$

$$\boxed{a_0 = 0}$$

$$b_n = 0$$

$$\text{Now for } a_n = \frac{1}{2} \int_{-1}^1 f(x) \frac{\cos n\pi x}{1} dx$$

$$= \frac{1}{2} \int_{-1}^1 (2x+1) \cos n\pi x dx$$

$$a_0 = \frac{1}{2} \int_{-1}^1 (2x+1) \cos 0 dx$$

$$= \frac{1}{2} \left[\frac{1}{4} + \frac{1}{2} + (1+1) + -\frac{1}{4} + \frac{1}{2} \right]$$

$$a_0 = 2 \int_0^1 (2x+1) \cos n\pi x dx$$

$$= \frac{1}{2} \left[\frac{1}{4} + \frac{1}{2} + (1+1) + -\frac{1}{4} + \frac{1}{2} \right] = 1$$

$$a_0 = 2 \left[\left[(2x+1) \frac{\sin n\pi x}{n\pi} \right]_0^1 - \int_0^1 2 \frac{\sin n\pi x}{n\pi} dx \right]$$

$$a_0 = 2 \left[\frac{1 \cdot \sin n\pi}{n\pi} - 0 + \left(\frac{2 \cos n\pi x}{n^2 \pi^2} \right)' \right]$$

$$a_0 = 4 \left[\frac{\cos n\pi x}{n^2 \pi^2} \right]_0^1 = \frac{4}{n^2 \pi^2} [\cos n\pi - \cos(0)]$$

$$a_0 = \frac{4}{n^2 \pi^2} [(-1)^n - 1] \quad \begin{cases} -\frac{8}{n^2 \pi^2} ; n \in \text{odd} \\ 0 ; n \in \text{even} \end{cases}$$

So the series is

$$f(x) = -1 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [(-1)^n - 1] \cos n\pi x$$

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$$f(x) = -1 - \frac{8}{\pi^2} \left[\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \frac{\cos 5\pi x}{5^2} + \dots \right]$$

* Parseval's Theorem:-

Let $f(x) \rightarrow$ Periodic function
(Period $\Rightarrow 2l$)

then

$$\frac{1}{2l} \int_c^{c+2l} [f(x)]^2 dx = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

where,

$$a_0 = \frac{1}{2l} \int_c^{c+2l} f(x) dx$$

$$b_n = \frac{1}{l} \int_c^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$a_n = \frac{1}{l} \int_c^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

Q Find the Fourier series for $f(x) = x^2$ in $(-\pi, \pi)$

Hence using Parseval's theorem: prove that -

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

(RTU-2013 & 24)

Solⁿ from the previous question - $(c \rightarrow -l)$

$$a_0 = \frac{\pi^2}{3}$$

$$a_n = \frac{4}{n^2} \cos n\pi = \frac{4}{n^2} (-1)^n$$

$$b_n = 0$$

So Fourier series

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos nx$$

$$x^2 = \frac{\pi^2}{3} - \frac{4}{1^2} \left[\frac{\cos 4}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \dots \right]$$

Now for Parseval's

$$\frac{1}{2\pi} \int_c^{c+2\pi} [f(x)]^2 dx = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} [a_n^2 + b_n^2]$$

$$(d=x)$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} (x^2)^2 dx = \left(\frac{\pi^2}{3}\right)^2 + \frac{1}{2} \sum_{n=1}^{\infty} \left[\left(\frac{4}{n^2} (-1)^n\right)^2 + 0^2 \right]$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{\pi^4}{9} + \frac{1}{2} \left[\sum_{n=1}^{\infty} \frac{16}{n^4} \right]$$

$$\frac{1}{2\pi} \left[\frac{x^5}{5} \right]_{-\pi}^{\pi} = \frac{\pi^4}{9} + 8 \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \right]$$

$$\frac{1}{2\pi} \left[\frac{\pi^5}{5} + \frac{\pi^5}{5} \right] = \frac{\pi^4}{9} + 8 \left(\sum_{n=1}^{\infty} \frac{1}{n^4} \right)$$

$$\frac{\pi^4}{5} = \frac{\pi^4}{9} + 8 \left(\sum_{n=1}^{\infty} \frac{1}{n^4} \right)$$

$$\pi^4 \left(\frac{1}{5} - \frac{1}{9} \right) = 8 \left(\sum_{n=1}^{\infty} \frac{1}{n^4} \right)$$

$$\pi^4 \left(\frac{4}{45} \right) = 8 \left(\sum_{n=1}^{\infty} \frac{1}{n^4} \right)$$

$$\boxed{\frac{\pi^4}{90} = \sum_{n=1}^{\infty} \frac{1}{n^4}}$$

Hence Proved